

InverPINN

Source-strength bias in physics-informed pollution inversion

How reliably can physics-informed neural networks recover a pollution source from sparse concentration measurements?

Khadis Aidyn

Sparse measurements, unknown source

Sensors observe concentration after transport.
The unknowns are source location x_s , y_s and peak intensity Q .

Almaty motivates the question; no real Almaty measurements enter this study.

20 fixed sensors.

Known transport physics

$$C_t + u C_x + v C_y = D(C_{xx} + C_{yy}) + QG$$

Wind carries concentration; diffusion spreads it; the source adds it.

One Gaussian, known width and transport. Zero initial and boundary concentration.

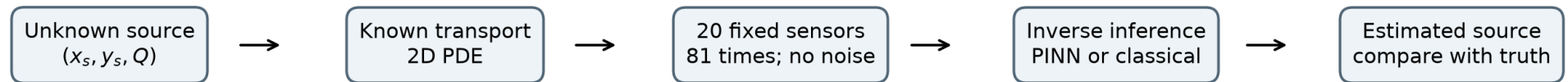
$u=0.35$, $v=-0.15$, $D=0.005$, $\sigma=0.08$.

Neural field and source inference

A PyTorch network represents $C(x,y,t)$.

Sensor loss fits observations; PDE residual couples the field to the source.

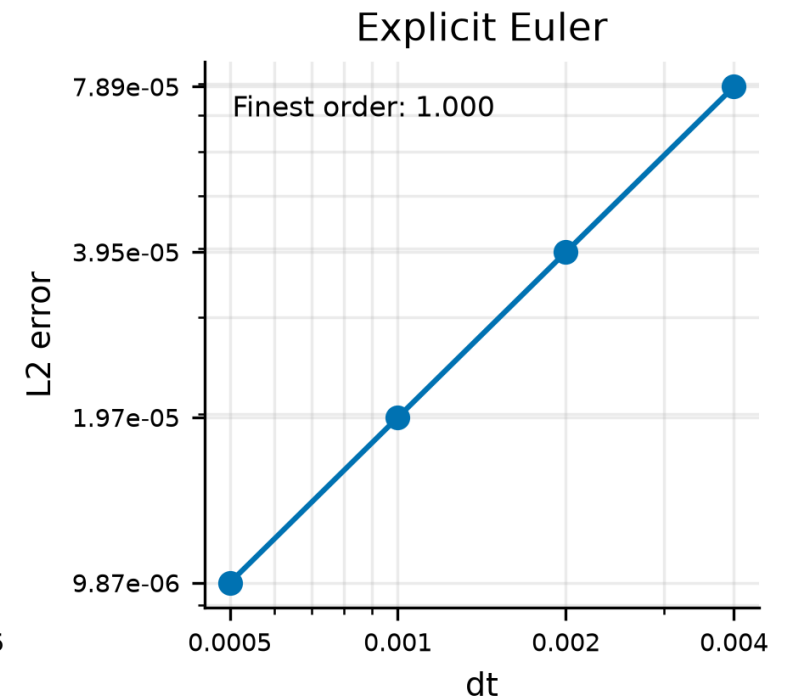
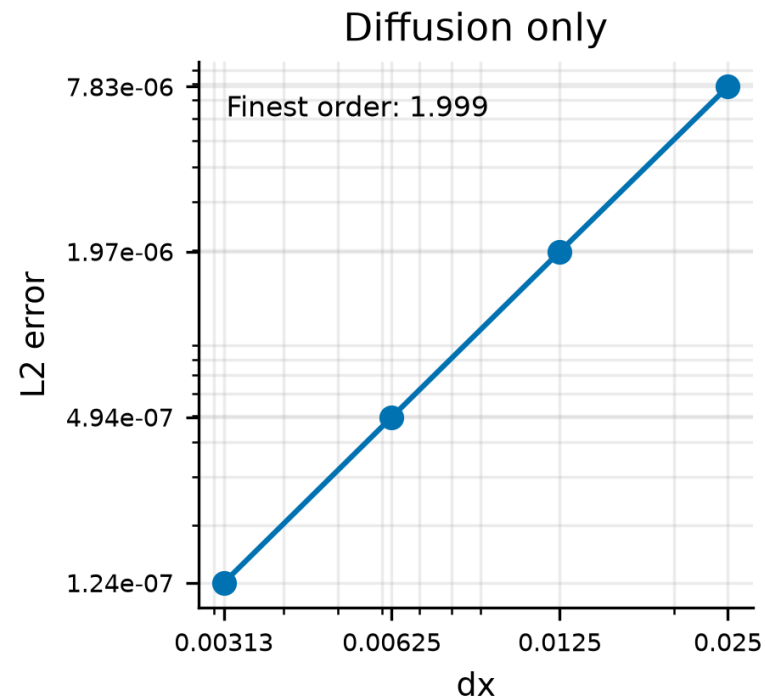
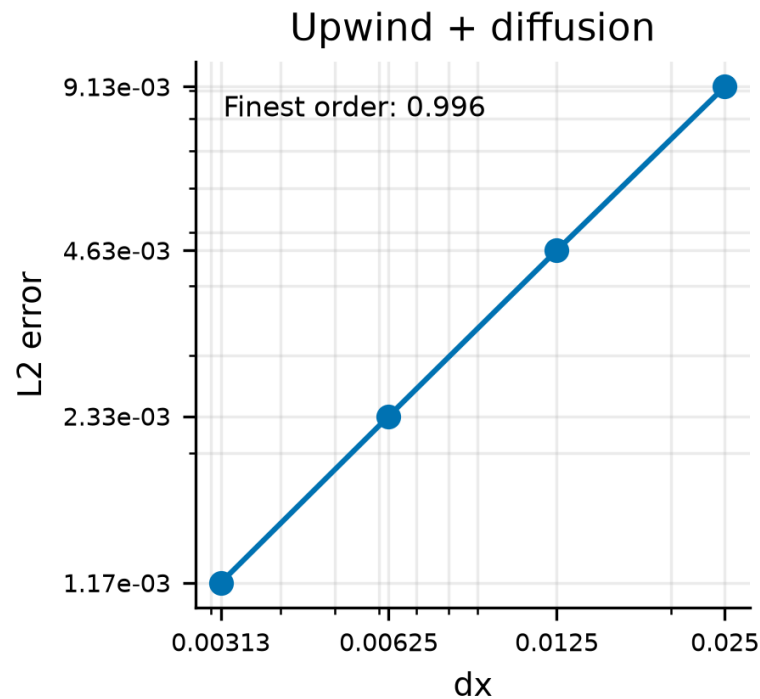
Hard constraints enforce nonnegative concentration and exact zero IC/BC.



Final blind confirmation: J1 23/30 < required 27/30 | Classical 30/30
Truth is available to evaluation, not fitting.

Reference quality and blind evaluation

641×641 numerical reference · 30 fresh final scenarios
Frozen methods and thresholds; all failures retained



Early failure and persistent Q bias

Earlier B0 held-out test: 14/30 recovered

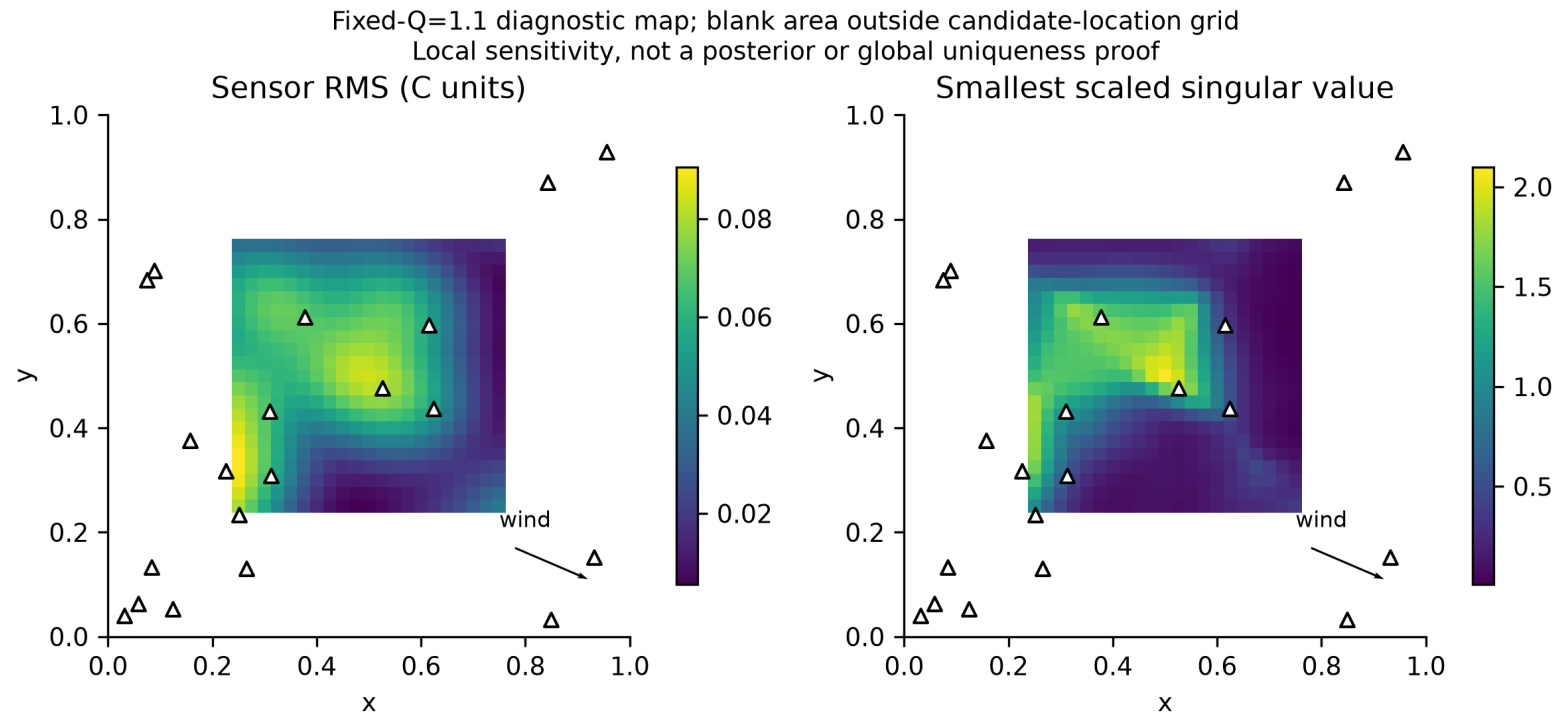
Median negative-prediction fraction: 35.424%

Hard constraints removed physical violations, but strength bias remained.

Historical B0 median negativity: 35.424%.

Classical inversion exposed the recovery gap

Classical inversion recovered 30/30 final cases on identical observations.
The tested observations supported better recovery than the PINN achieved.



J1 profiles source amplitude analytically

Residual: $R0 - QG$

$$Q^* = \text{sum}(w_i G_i R0_i) / \text{sum}(w_i G_i^2)$$

Q leaves Adam's parameter list. The field and location remain trainable.

This minimizes the sampled PDE objective, not observation mismatch through a solver.

Q is profiled each update, plus a fixed terminal profile.

J1 failed final confirmation

23/30 = 76.67%; required 27/30 = 90%
28 Q underestimates / 2 overestimates

Error columns report medians across all 30 fresh scenarios

Method	Recovery	Localization	Q error	Field L2
J1 PINN	23/30	0.005574	5.975%	7.423%
B_revised	21/30	0.009186	9.319%	11.048%
Classical	30/30	0.001111	1.218%	1.507%

Joint recovery: localization ≤ 0.08 AND relative Q error $\leq 20\%$

Physical validity did not guarantee recovery

All seven J1 joint failures missed the Q criterion.
Development stopped after failed confirmation.
Noise, uncertain weather and real Almaty attribution remain unvalidated.

final_blind_v1_030: failure; t=0.8
Full-grid scores: localization=0.0841, Q error=62.2%, L2=69.7%

